ZOMATO
Sales & Customer Analytics
Dashboard Project

Business Intelligence · SQL · Power BI · DAX

End-to-End Analytics Workflow
Raw Data → SQL Cleaning & Analysis → Power BI Dashboard

Prepared by: Analytics Team
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1. Project Overview

The Zomato Sales & Customer Analytics Dashboard project was developed to analyze restaurant operations, customer behavior, and sales performance using SQL and Power BI. The project transforms raw business data into meaningful insights through data cleaning, SQL analysis, KPI generation, and interactive dashboard development.

The project follows a real-world business intelligence workflow used in modern analytics environments. The main objective is to identify business trends, customer behavior patterns, restaurant distribution insights, and operational metrics that support business decision-making.

2. Project Objectives

The main objectives of this project are:

- Analyze overall sales and order performance

- Identify customer retention and repeat customer behavior
- Monitor average customer spending patterns
- Analyze restaurant distribution across cities
- Study cuisine popularity and restaurant ratings
- Build interactive dashboards for business storytelling
- Create professional analytical reports using SQL and Power BI

3. Tools & Technologies Used

Tool / Technology	Purpose
PostgreSQL	Database management and SQL analysis
SQL	Data cleaning, querying, aggregation, KPI analysis
Power BI	Dashboard development and visualization
DAX	Calculated measures and KPIs
CSV Dataset	Source data files

4. Dataset Information

The project uses multiple datasets related to restaurant operations, customer orders, and restaurant details.

Main Tables Used

Table	Description	Key Columns
orders_cleaned	Order-level transactional information	order_date, sales_amount, sales_qty, user_id, r_id
restaurants_cleaned	Restaurant-related information	restaurant_id, restaurant_name, city, cuisine, rating, cost

users	Customer-related information	user_id, customer details

5. Data Import & Preparation

The raw CSV files were imported into PostgreSQL for structured analysis and processing. Initial validation checks were performed to ensure data accuracy before dashboard development.

Validation Checks Performed

- Column validation
- Data type verification
- Row count validation
- Null value identification
- Duplicate checking
- Relationship verification

6. Data Cleaning Process

Data cleaning was performed using SQL to improve data quality and analytical accuracy. The following cleaning operations were completed:

- Removed duplicate records
- Standardized inconsistent values
- Cleaned currency formatting issues
- Validated city and cuisine information
- Handled null and missing values
- Corrected formatting inconsistencies
- Verified relationships between tables
- Created cleaned analytical tables

Cleaned Tables Created

- orders_cleaned
- restaurants_cleaned

This step improved dashboard reliability and analytical consistency.

7. SQL Analysis

SQL was used as the primary analytical layer before visualization in Power BI. The SQL analysis process included data aggregation, customer analysis, revenue analysis, restaurant performance analysis, KPI calculation, and trend analysis.

SQL Concepts Used

SQL Concept	Application
SELECT Statements	Data retrieval and projection
WHERE Filtering	Conditional data extraction
GROUP BY / ORDER BY	Data aggregation and sorting
Aggregate Functions	SUM, COUNT, AVG, MAX, MIN
JOIN Operations	Multi-table data combination
DISTINCTCOUNT Logic	Unique customer/order counting
CASE Statements	Conditional business logic

8. Key Performance Indicators (KPIs)

Sales KPIs

Total Revenue	Total Orders	Avg Order Value	Revenue Trend

Customer KPIs

Total Customers	Repeat Customers	Retention %	Avg Spend

Restaurant KPIs

Total Restaurants	Total Cities	Total Cuisines	Avg Rating

9. Data Modeling in Power BI

The cleaned SQL tables were imported into Power BI for dashboard development. Relationships were created between orders_cleaned, restaurants_cleaned, and users tables.

Data Model Design Goals

- Maintain relational integrity across all tables
- Improve dashboard performance through optimized relationships
- Support cross-filtering across dimensions
- Enable business analysis across multiple analytical dimensions

10. Dashboard Development

Three interactive dashboard pages were created in Power BI, each serving a distinct analytical purpose.

Dashboard Page 1 — Sales & Customer Overview

Purpose: Provide executive-level business performance monitoring.

- Total Revenue KPI card
- Total Orders KPI card
- Active Customers KPI card
- Average Order Value KPI card
- Revenue Trend Analysis (line/bar chart)
- Revenue by City (geographic breakdown)
- Interactive Filters and Slicers

Dashboard Page 2 — Customer Analytics

Purpose: Analyze customer behavior and spending patterns.

- Total Customers & Repeat Customers
- Customer Retention Percentage
- Average Customer Spend
- Customer Segmentation visuals
- Monthly Active Customers Trend
- Top Customers by Spend
- Customer Detail Table

Dashboard Page 3 — Restaurant Analytics

Purpose: Analyze restaurant operations and cuisine distribution.

- Total Restaurants & Average Cost
- Total Cities & Total Cuisines
- Top Cities by Restaurant Count
- Cuisine Distribution chart
- Average Cost by City

- Restaurant Rating Distribution

11. Business Insights Generated

Customer Insights

- A significant percentage of customers are repeat customers
- High-value customers contribute a major share of total revenue
- Customer retention is above 50%, indicating strong platform engagement

Restaurant Insights

- Restaurant concentration is higher in selected major cities
- Certain cuisines dominate the platform distribution
- Most restaurants are clustered around higher rating ranges

Sales Insights

- Revenue trends vary across years — useful for forecasting
- Average order value helps measure and benchmark customer spending behavior
- City-level analysis identifies high-performing regions for expansion

12. Why SQL Was Used Before Power BI

SQL analysis was performed before dashboard creation to simulate a real-world professional analytics workflow:

1. Improve dashboard performance through pre-aggregated data
2. Clean raw data efficiently using structured SQL logic
3. Apply business logic before visualization layer
4. Create reusable analytical tables for multiple reports

5. Simulate real-world industry BI workflow standards
6. Simplify Power BI dashboard development complexity

Raw Data → SQL Cleaning & Analysis → Power BI Dashboard

This workflow reflects real-world business intelligence practices commonly used in professional organizations.

13. Challenges Faced

Several technical and analytical challenges were encountered and resolved during the project:

Challenge	Resolution Approach
Data import issues	Validated file formats and encoding before import
Incorrect aggregation in Power BI	Applied proper DAX measures and filter context
Data cleaning inconsistencies	Used SQL validation queries and cross-checks
Filter context issues	Rebuilt data model with correct relationships
Visualization formatting	Applied consistent theme and formatting rules
Distinct count calculation	Used DISTINCTCOUNT DAX function with validated logic

14. Project Outcomes & Skills Demonstrated

This project demonstrates the following professional skills:

Technical Skills	Analytical Skills
SQL querying and analysis	KPI creation and business analysis
Data cleaning and preprocessing	Business storytelling with data

Power BI dashboard development	Data visualization best practices
DAX measure creation	Analytical problem-solving
Data modeling (relational)	End-to-end BI workflow execution

15. Conclusion

The Zomato Sales & Customer Analytics Dashboard project successfully transforms raw business data into actionable insights using SQL and Power BI. The project demonstrates an end-to-end analytics workflow including:

- Data cleaning and preprocessing
- SQL analysis and KPI generation
- Dashboard development and visualization
- Business storytelling and insight communication

This project reflects a real-world business intelligence process used in professional analytics environments and highlights practical analytical and visualization skills.

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